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Lightspeed Core LCORE-1311

# Conversation History Summarization (Compaction)

## Details

| | | | |
|---|---|---|---|
| Type: | Feature | Resolution: | Unresolved |
| Priority: | Undefined | Fix Version/s: | Q2CY26 |
| Affects Version/s: | None | | |
| Component/s: | None | | |
| Labels: | ols-lcore-migration | | |
| Blocked: | False | | |
| Blocked Reason: | None | | |
| Ready: | False | | |
| SFDC Cases Links: | | | |
| SFDC Cases Open: | 0 | | |
| SFDC Cases Counter: | 0 | | |
| Intelligence Requested: | False | | |
| Market: | Unclassified | | |

## Description

# LCORE Feature Request: Conversation History Summarization

## Summary

Implement intelligent conversation history summarization to maintain context when conversations approach the model's context window limit, preventing loss of important context while enabling long-running conversations.

## Background

Related ticket: OLS-2500 - [Review] : Summarize conversation when context limit is reached

## User Story

> As an LCS user, I want the conversation history to be intelligently summarized when it exceeds the context window limit, so that the assistant maintains an accurate understanding of previous interactions without losing important context.

## Problem Statement

Currently, neither lightspeed-stack nor Llama Stack implements context management for long conversations:

- When context window is exceeded, requests fail with HTTP 413 (Prompt Too Long)
- No automatic truncation or summarization exists
- The `truncated` field in responses is hardcoded to `False` with TODO comments

## Technical Analysis

### Current State

| Component | Behavior |
|---|---|
| **lightspeed-stack** | Returns `PromptTooLongResponse` (413) on overflow. `truncated` field not implemented. |
| **Llama Stack** | Loads ALL conversation history via Conversations API + ResponsesStore. No truncation/summarization. |
| **OpenAI Responses API** | Has `truncation` parameter (auto/disabled) AND `compact` endpoint for summarization |

### OpenAI API Spec (Reference Implementation)

OpenAI's Responses API includes two relevant features:

**1. `truncation` parameter** on create response:

- `auto`: Drops items from beginning of conversation to fit context window
- `disabled` (default): Fails with 400 error if context exceeded

**2. `compact` endpoint** (`POST /v1/responses/compact`):

- Runs compaction/summarization pass over conversation
- Returns compacted response object with encrypted/opaque items

**Neither feature is currently implemented in Llama Stack.**

### Llama Stack Storage Architecture

Llama Stack uses two storage mechanisms for conversation history:

**1. Conversations API** (`conversations_api.list_items()`)

- Stores `ConversationItem` objects (high-level: messages, tool calls)
- Used for listing/managing conversations

**2. ResponsesStore** (`responses_store.get_conversation_messages()`)

- Stores raw `OpenAIMessageParam` list (what actually goes to LLM)
- Stored in `conversation_messages` table
- Source of truth for building LLM context

Both mechanisms store full history without any summarization or truncation.

## Impact Analysis

| Area | Impact |
|---|---|
| **Transcripts** | No impact - captures individual Q/A pairs, not conversation history |
| **Conversation Cache** | TBD - needs schema changes to store summary metadata |
| **Llama Stack** | Depends on implementation location |

### Transcripts Detail

Transcripts store individual Q/A pairs per turn. The `truncated` field already exists to flag when truncation occurred. If summarization happens, transcripts would still capture the original query/response for that turn - this is the intended behavior.

### Conversation Cache Detail

If implemented in lightspeed-stack, the conversation cache would need schema changes.

## Design Considerations

### Key Principle: Decouple Storage from LLM Context

**Storage and LLM context should be decoupled:**

- Full history preserved (for UI/audit/replay)
- Summary used only when building LLM context
- User sees all messages; AI gets summarized context for older turns

This follows the pattern used by tools like Cursor/Claude - you can scroll up and see your full conversation, but the AI may only have summarized context of older turns.

### Summarization Trigger

Summarization should be token-based, not turn-based (turn sizes vary significantly).

### Incremental Summarization

Summary should be computed once, stored, and reused - not recomputed on every request.

## Open Questions

1. Should this be implemented in Llama Stack (where conversation history is managed) or lightspeed-stack?
2. If Llama Stack, what's the timeline/feasibility for upstream contribution?
3. How should summary storage interact with existing conversation cache schema?
4. What summarization prompt/strategy provides best context preservation?
5. How to handle model-specific context window sizes (configurable vs auto-detected)?

## Acceptance Criteria

1. When conversation history approaches context window limit, older messages are summarized (not just truncated)
2. Summary preserves clear attribution of actions (user vs assistant)
3. Summarization threshold is configurable or auto-determined based on model context window
4. Summarization is incremental (summary updated, not recomputed from scratch)
5. Full conversation history remains accessible (UI/audit) - only LLM context uses summary
6. Assistant correctly recalls and references prior context after summarization

## References

- OLS-2500: Original OLS ticket
- OpenAI Responses API
- OpenAI Compact endpoint
- OpenAI Conversation State Guide

## Attachments

Drop files to attach, or browse.

## Issue Links

**is blocked by**

LCORE-1314 [Spike] Finalize scope of the feature request IN PROGRESS

## Easy Agile Planning Poker

Live updates Quick tour

Vote

## Child issues

No child issues.

## Activity

**Newest first**

You can now pin up to five comments to highlight important information. Pinned comments will appear above all other comments, so they're easy to find.
Got it Learn more about pinned comments

Anxhela Coba added a comment - 2026/02/11 4:29 PM

Is the goal of the conversation history summarization task more about preventing missing information? In fine-tuning when we compress models from larger to more domain-specific they can be prone to catastrophic forgetting. I wonder if we can explore for this.. techniques like SVD ( singular vector decomposition) for this task.

ex. if we can take the embedding of the history we can apply SVD or some similar technique to help "summarize"

Pin

Ondrej Metelka added a comment - 2026/02/09 4:07 PM

Note: if it is implemented on the provider/openai API level - then this feature is locked for providers conforming to these particular endpoints.

Pin

## People

Assignee:

Unassigned

Assign to me

Reporter:

Ondrej Metelka

Votes:

0 Vote for this issue

Watchers:

3 Start watching this issue

**Dates**

Created:
2026/02/09 3:45 PM

Updated:
2026/03/10 3:13 PM

**Agile**